

LLM judges against human experts

Measuring faithfulness judgment on difficult cases across three deployment tiers

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Research question

How much LLM-judge agreement with human experts survives chance correction on difficult faithfulness cases, and does the remainder support a production release gate?

PART ONE

How much of the agreement is real once luck is removed.

PART TWO

Whether what is left is enough to block a release.

Background

An LLM judge scores model output at scale, and its verdict decides what ships.

The practice rests on one published number: about 80 percent agreement with human experts (Zheng et al., 2023).

That number is raw agreement, uncorrected for luck.

If nine summaries in ten are bad, a judge that flags everything without reading scores 90 percent.

Cohen's kappa reports only the share of agreement that luck did not already hand over.

Data

FaithBench (Bao et al., 2025). Summaries with expert hallucination labels, collected because automatic detectors disagreed on them.

750

summaries in the corpus,
split 262 development
and 488 test

488

test items, read once, on
9 August 2026

68.6%

of test items are
unfaithful under the
primary scheme

82.0%

agreement between two
human experts on the
same binary task

Four levels, two classes

The human annotators used four severity levels. A release gate has two states. The mapping between them is a decision, so I took the benchmark authors' own.

HUMAN LEVEL	WHAT IT MEANS	TEST ITEMS	THIS STUDY
Consistent	Nothing unsupported by the source	113	Faithful
Benign	A hallucination, but one a reader accepts	40	Faithful
Questionable	Unclear; the annotators themselves disagreed	45	Unfaithful
Unwanted	A hallucination that misleads the reader	290	Unfaithful

WHY THE LINE SITS THERE

It is the split in FaithBench's own released binarisation code, so the numbers stay comparable. The two stricter splits are reported as sensitivity checks.

WHAT THE JUDGE IS ASKED

Faithful or unfaithful, and nothing else. The four levels stay on the human side only; a gate makes a binary decision.

The three judges

TIER	MODEL	DATA LEAVES THE MACHINE	COST	MEDIAN LATENCY
Paid API	claude-opus-5	Yes	22.21 USD	3.9 s
Hosted open weights	gpt-oss:120b-cloud	Yes	Free tier	5.0 s
Local, on the laptop	gpt-oss:20b	No	Free	15.2 s

Three judges, 750 items, three repeats each. 7,012 judgments in the store, 4,392 of them on the test split.

Measurement discipline

01

One frozen question

Prompt v0.2, hash 3b14635aa1c3eedc, carried on every judgment row in the store.

02

Predictions before data

Fourteen numbered quantitative predictions, committed to version control before the test half was read.

03

One reading of the test split

488 items, opened once, after the rubric and the analysis code were frozen with file hashes.

Ten predictions held, two held in part, one weakened to its margin, and one was overturned. The overturned one is in the paper, marked as overturned.

Live demo

One judgment, end to end, on the local model. No API key, no network.

01

The instrument: the rubric and its hash

02

The case: source and summary, read aloud

03

The judge runs, about fifteen seconds

04

Why the judge and the experts disagreed

05

The numbers, re-derived from the store

faithbench-judge-study · .venv/bin/python demo.py --gate

Headline results

JUDGE	RAW AGREEMENT	COHEN'S KAPPA, 95% CI	DEFLATION	CATCHES UNFAITHFUL	PASSES FAITHFUL
Tier 1, paid API	46.6%	0.129 [0.084, 0.175]	33.7 pp	25%	93%
Tier 2, hosted	51.4%	0.162 [0.109, 0.214]	35.2 pp	35%	87%
Tier 3, local	47.3%	0.131 [0.083, 0.178]	34.1 pp	27%	92%
Flag everything, read nothing	68.6%	0.000	—	100%	0%
Two human experts	82.0%	—	—	—	—

Test split, 488 items, scheme U+Q. Bootstrap intervals resampled over items, 2,000 resamples.

Three findings

34 points

Chance correction removes 33.7 to 35.2 points on every tier. Norman et al. (2026) measured 33 to 41 points on a different benchmark and a different set of judges.

68.6%

A rule that flags every summary without reading it scores higher than all three judges, which reached 46.6, 51.4 and 47.3 percent.

22 USD

No tier is distinguishable from another. All three intervals overlap across the band 0.109 to 0.175, and the paid tier is nominally the lowest of the three.

Discussion and limits

WHAT SURPRISED ME

No tier was distinguishable from any other, and the model running free on my laptop kept up with the paid frontier API.

TIER 1, PAID API	TIER 2, HOSTED	TIER 3, LOCAL LAPTOP
0.129	0.162	0.131
[0.084, 0.175] · 22.21 USD	[0.109, 0.214] · free tier	[0.083, 0.178] · free

All three intervals overlap across the band 0.109 to 0.175, so the data cannot tell the tiers apart. The paid tier is nominally the lowest of the three.

Limits: FaithBench items were selected because detectors disagreed on them, so this is a worst case. One benchmark, English only, faithfulness only. Three tiers are three points, not a trend.

Conclusion and outlook

Between 13 and 16 percent of the available agreement survives chance correction, and it does not support a production gate.

THE ANSWER IN TWO LINES

Raw agreement of 46.6 to 51.4 percent falls to a kappa of 0.129 to 0.162 once chance is removed.

Every judge agrees with the human labels less often than a rule that reads nothing.

Never report raw agreement on its own. Correcting for chance removed 33.7 to 35.2 points of it on every tier.